ACCENTURE 2026–2027 HIRING · TECHNICAL ASSESSMENT · SECTION C

Networking, Security & Cloud Complete Guide

> ≈ 14 of 45 questions · the largest recall section · highest-scoring if revised_

Everything for Section C, from zero: network basics, the OSI and TCP/IP models, devices, IP addressing and subnetting, TCP/UDP, every key protocol and port, cybersecurity (CIA, cryptography, authentication, malware, attacks, defences), cloud computing and DevOps (SDLC, Agile, CI/CD, Git, Docker, Kubernetes). Includes 64 practice MCQs.

🌐 Networking | 🔐 Security | ☁️ Cloud | ⚙️ DevOps

≈ 14	40+	64	✔
Questions in the test	Ports and protocols	Practice MCQs	Subnet answers computed in code

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Edition: 2026–2027

INSIDE THIS GUIDE

Contents

00 What's tested and how to prepare

N1 Network basics, types and topologies

N2 OSI and TCP/IP models

N3 Network devices and domains

N4 IP addressing, subnetting and IPv6

N5 Transport: TCP vs UDP and ports

N6 Application protocols (HTTP, DNS, DHCP...)

N7 Routing, switching and Wi-Fi

S1 Security fundamentals

S2 Cryptography and PKI

S3 Authentication and access

S4 Malware and attacks

S5 Network defences and testing

K1 Cloud computing and service models

K2 Virtualisation, scaling and storage

K3 SDLC, Agile and DevOps

K4 Git, Docker and Kubernetes

✓ Practice MCQs, answer keys and cheat sheet

KNOW THE SECTION

00 What's Tested & How to Prepare

Section C is the **largest section**, with about 14 of the 45 questions. The questions are mostly **direct recall**: which layer, which port, which attack, which cloud model. That makes it the best place to score heavily if you revise the tables in this guide.

Area	Typical questions	Weight*
Networking	OSI layer of a device or protocol · port numbers · TCP vs UDP · IP class or private range · subnet host count · DNS/DHCP steps · HTTP status codes	~40%
Security	CIA triad · symmetric vs asymmetric · hashing · malware types · phishing variants · SQLi/XSS · firewall/IDS/IPS/VPN	~30%
Cloud & DevOps	IaaS/PaaS/SaaS examples · deployment models · virtualisation · scaling · CI/CD · Git commands · Docker terms	~30%

*An approximate split based on the published syllabus, not an official breakdown.

How to prepare

Learn the **tables**, because they hold about 80% of the answers. Use flashcards for ports and layers. For subnetting, learn the 3 formulas in N4 and practise until a /27 or /26 question takes under 30 seconds.

🌐 PART 1 · NETWORKING

Computer Networks

How devices connect and communicate: the models, the hardware, the addresses and the protocols. About 40% of Section C.

N

N1

THE BASICS

Network Basics, Types & Topologies

Network: two or more devices (nodes) connected to share data and resources. A **protocol** is an agreed set of rules for communication. **Bandwidth** = maximum data rate (bps). **Throughput** = the rate actually achieved. **Latency** = delay (ms). 1 Byte = 8 bits, so 100 **Mbps** ≈ 12.5 **MB/s**.

Type	Coverage	Example
PAN	About 10 m around a person	Bluetooth earbuds
LAN	A room, building or campus	Office Wi-Fi or Ethernet
MAN	A city	A city cable network
WAN	A country or the world	The **Internet** (the largest WAN)

Mode	Direction	Example
Simplex	One way only	Keyboard → PC, TV broadcast
Half duplex	Both ways, one at a time	Walkie-talkie
Full duplex	Both ways at once	Phone call

Topology	Layout	Pros	Cons
Bus	All devices on one backbone cable	Cheap, simple	The backbone is a single point of failure, and collisions are common
Star	All devices connect to a central switch or hub	Easy to manage. One broken cable affects only one device	The central device is a single point of failure. The most common LAN layout
Ring	Each device connects to two neighbours in a loop	Orderly (token passing)	One break can stop the ring (unless it's a dual ring)
Mesh	Every device links to every other (full mesh)	Very reliable, with many paths	Expensive: needs **n(n−1)/2** links
Tree	Hierarchical, with stars attached to a backbone	Scalable	Depends on the root
Hybrid	A mix of the above	Flexible	Complex

Transmission media

Twisted pair (Cat5e/Cat6, RJ-45, most LANs) · **Coaxial** (cable TV) · **Fibre optic** (light, fastest, long distance, immune to interference. Single-mode = long distance, multi-mode = short) · **Wireless** (radio, microwave, infrared).

Architectures

Client–server: central servers provide services (web, mail). **Peer-to-peer**: every node is both client and server (BitTorrent).

Switching methods

Circuit switching: a dedicated path for the whole session (old telephone network). **Packet switching**: data is split into packets that may take different routes (the internet).

N2

THE MOST-TESTED MODEL

OSI & TCP/IP Models

7	6	5	4	3	2	1
Application	Presentation	Session	Transport	Network	Data Link	Physical

Mnemonic: top to bottom, "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing". Bottom to top, "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way".

#	Layer	Responsibility	Protocols / standards	Devices	PDU	Address
7	**Application**	Network services for applications and users	HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS, DHCP, SSH, Telnet, SNMP	Gateway, L7 firewall	Data	—
6	**Presentation**	Translation, encryption, compression	SSL/TLS*, JPEG, GIF, MPEG, ASCII	—	Data	—
5	**Session**	Set up, manage and end sessions; checkpoints	NetBIOS, RPC, PPTP	—	Data	—
4	**Transport**	End-to-end delivery, segmentation, flow and error control, **ports**	TCP, UDP	Load balancer (L4)	Segment / datagram	Port
3	**Network**	Logical addressing, **routing**, path selection	IP (v4/v6), ICMP, IPsec, OSPF, RIP	**Router**, L3 switch	Packet	IP address
2	**Data Link**	Node-to-node delivery, framing, MAC, error detection (CRC)	Ethernet (802.3), Wi-Fi (802.11), PPP, ARP*	**Switch**, bridge, NIC	Frame	MAC address
1	**Physical**	Bits over the medium: voltage, light, radio	Cables, RJ-45, USB, Bluetooth radio	**Hub**, repeater, modem	Bits	—

*TLS is usually placed at Layer 6 (some sources say 4–7). ARP works between Layers 2 and 3.

Encapsulation: data going down the stack

Data → + TCP header = Segment →

+ IP header = Packet →

+ MAC header/trailer = Frame → Bits

The receiver strips each header in reverse order (**de-encapsulation**).

TCP/IP layer	OSI equivalent
Application	7 + 6 + 5
Transport	4
Internet	3
Network Access / Link	2 + 1

OSI vs TCP/IP: OSI is a 7-layer **reference/teaching** model. TCP/IP is a 4-layer **practical** model that the internet actually uses. (Some books describe TCP/IP with 5 layers by splitting Link into Data Link and Physical.)

N3

THE HARDWARE

Network Devices & Domains

Device	Layer	What it does
Repeater	1	Regenerates the signal to extend the distance
Hub	1	Broadcasts every bit to **all** ports. No intelligence. One collision domain
Bridge	2	Connects two LAN segments and filters by MAC
Switch	2	Forwards frames only to the right port using a **MAC address table**. Each port is its own collision domain
Router	3	Connects **different networks** and forwards packets by **IP** using a routing table. Separates broadcast domains
Gateway	All / 7	Connects networks that use **different protocols**. The "default gateway" is the router leading out of your LAN
Modem	1	Modulates and demodulates between digital data and analog signals (DSL, cable)
Access point	2	Connects wireless devices to a wired LAN
Firewall	3–7	Filters traffic by rules (see S5)
Load balancer	4 / 7	Spreads traffic across several servers
NIC	1–2	Network interface card. It holds the MAC address

Collision vs broadcast domain: a **hub** has 1 collision domain and 1 broadcast domain. A **switch** has a collision domain **per port** but 1 broadcast domain (unless VLANs are used). A **router** has a separate broadcast domain **per interface**.

N4

ADDRESSES

IP Addressing, Subnetting & IPv6

IPv4 = 32 bits written as 4 **octets** in dotted decimal (each 0–255), for example `192.168.1.10`. Every address has a **network part** and a **host part**, split by the **subnet mask** (`255.255.255.0`, or **/24** in CIDR notation, meaning 24 network bits).

Class	1st octet	Default mask	Use
A	1 – 126	/8 (255.0.0.0)	Very large networks
B	128 – 191	/16 (255.255.0.0)	Medium
C	192 – 223	/24 (255.255.255.0)	Small
D	224 – 239	—	Multicast
E	240 – 255	—	Experimental

Special address	Meaning
10.0.0.0/8	Private (Class A range)
172.16.0.0 – 172.31.255.255	Private (/12)
192.168.0.0/16	Private (home routers)
127.0.0.0/8 (127.0.0.1)	Loopback (localhost)
169.254.0.0/16	APIPA: DHCP failed
0.0.0.0 / 255.255.255.255	"This host" / limited broadcast

Subnetting in 3 formulas

1. **Host bits** h = 32 − prefix → **usable hosts** = 2^h − 2 (minus the network and broadcast addresses).
2. **Block size** = 256 − (the mask's last non-255 octet) = 2^h in the last octet. Subnets start at multiples of the block size.
3. For any host: **network** = the block start at or below the host · **broadcast** = next block start − 1 · **usable hosts** = network + 1 up to broadcast − 1.

Worked example: 192.168.10.77/26. h = 6 → 62 hosts. The mask is 255.255.255.192, so the block is 64 and the blocks are .0, .64, .128, .192. 77 is in the **.64** block: network **192.168.10.64**, broadcast **.127**, hosts **.65–.126**.

Prefix	Mask	Block	Usable hosts	Subnets in a /24
/24	255.255.255.0	256	254	1
/25	255.255.255.128	128	126	2
/26	255.255.255.192	64	62	4
/27	255.255.255.224	32	30	8
/28	255.255.255.240	16	14	16
/29	255.255.255.248	8	6	32
/30	255.255.255.252	4	2 (point-to-point links)	64

IPv6

128 bits, written as 8 groups of 4 hex digits: `2001:0db8:0000:0000:0000:0000:0000:0001` → `2001:db8::1`. Leading zeros can be dropped, and **::** can be used **once**. There's **no broadcast** (it uses multicast and anycast instead). Loopback is `::1`.

NAT / PAT

NAT translates private IPs to a public IP at the router. **PAT** (NAT overload) lets many devices share **one** public IP by using different ports. It's why home networks work with one ISP address.

MAC address

48-bit physical (hardware) address, like `00:1A:2B:3C:4D:5E`. The first 24 bits are the **OUI** (manufacturer). **ARP** maps IP → MAC, and the old RARP maps MAC → IP.

Public vs private IP · static vs dynamic: a public IP is globally unique and routable on the internet. A private IP is reusable inside LANs and isn't routed on the internet. A **static** IP is fixed (servers). A **dynamic** IP is assigned by DHCP (laptops, phones). **Supernetting** or CIDR aggregation combines networks, such as two /24s into one /23.

N5

END-TO-END DELIVERY

Transport: TCP vs UDP & Ports

	TCP	UDP
Connection	**Connection-oriented**: 3-way handshake **SYN → SYN-ACK → ACK**	**Connectionless**
Reliability	Guaranteed delivery, acknowledgements, retransmission, **ordering**	Best effort: no ACKs, no ordering
Control	Flow control (sliding window) and congestion control	None
Header	20+ bytes	8 bytes
Speed	Slower	Faster, low latency
Used by	HTTP/HTTPS, FTP, SMTP, IMAP, SSH	DNS queries, DHCP, streaming, VoIP, gaming, TFTP, SNMP

Connection teardown uses 4 steps: FIN → ACK → FIN → ACK. **Port ranges**: well-known **0–1023** · registered 1024–49151 · dynamic/ephemeral 49152–65535. A **socket** = IP address + port number.

N6

LAYER 7

Application Protocols

Port	Protocol	Transport	Purpose	Port	Protocol	Transport	Purpose
20/21	FTP	TCP	File transfer (data/control)	110	POP3	TCP	Download mail
22	SSH / SFTP	TCP	Secure shell and file transfer	123	NTP	UDP	Time sync
23	Telnet	TCP	Remote login (plain text)	143	IMAP	TCP	Mail synced on the server
25	SMTP	TCP	Send mail (587 = submission)	161/162	SNMP	UDP	Network monitoring
53	DNS	UDP/TCP	Names → IP	389 / 636	LDAP / LDAPS	TCP	Directory services
67/68	DHCP	UDP	Auto IP (server/client)	443	HTTPS	TCP	Secure web (TLS)
69	TFTP	UDP	Trivial file transfer	993 / 995	IMAPS / POP3S	TCP	Secure mail
80	HTTP	TCP	Web	3306 / 1433	MySQL / MS SQL	TCP	Databases
88	Kerberos	TCP/UDP	Authentication tickets	3389	RDP	TCP	Remote desktop

HTTP / HTTPS

Request–response and **stateless** (cookies and sessions add state). Methods: **GET** (read), **POST** (create), **PUT** (replace), **PATCH** (partial update), **DELETE**, HEAD, OPTIONS. Versions: HTTP/1.1 (keep-alive) · HTTP/2 (multiplexing, binary) · HTTP/3 (over **QUIC**, which runs on **UDP**). **HTTPS** = HTTP over **TLS**, using a certificate plus encryption.

Code	Meaning
1xx	Informational
2xx	**Success**: 200 OK · 201 Created · 204 No Content
3xx	**Redirect**: 301 Moved Permanently · 302 Found · 304 Not Modified
4xx	**Client error**: 400 Bad Request · 401 Unauthorized · 403 Forbidden · 404 Not Found · 429 Too Many Requests
5xx	**Server error**: 500 Internal · 502 Bad Gateway · 503 Unavailable · 504 Gateway Timeout

DNS: how www.example.com is resolved

1. Check the browser cache, then the OS cache and hosts file.
2. Ask the **recursive resolver** (usually the ISP's, or 8.8.8.8).
3. The resolver asks a **root** server, which sends it to the **.com TLD** server, which sends it to the **authoritative** server for example.com.
4. The IP is returned and cached for the record's **TTL**.

Records: **A** (IPv4) · **AAAA** (IPv6) · **CNAME** (alias) · **MX** (mail) · **NS** (name server) · **PTR** (reverse lookup) · **TXT** (verification).

DHCP: DORA

1. **D**iscover: the client broadcasts "any DHCP server out there?"
2. **O**ffer: the server offers an IP address.
3. **R**equest: the client requests that offer.
4. **A**cknowledge: the server confirms. The client gets an IP, mask, gateway and DNS for a **lease** period.

ICMP carries `ping` (echo request/reply) and `traceroute` (TTL expiry). **Useful commands**: ipconfig/ifconfig, ping, tracert/traceroute, nslookup, netstat.

GETTING PACKETS AROUND

N7 Routing, Switching & Wi-Fi

Routing

Static routes are set manually (simple and predictable). **Dynamic** routes are learned automatically:

- **RIP**: distance-vector, metric = hop count, max 15 (16 = unreachable)
- **OSPF**: link-state, shortest path (Dijkstra), used inside organisations
- **BGP**: path-vector, routes **between** autonomous systems. It's the protocol of the internet.

The **default gateway** is where packets go when no specific route matches.

Switching concepts

A **MAC table** maps port ↔ MAC. A **VLAN** logically splits one switch into separate broadcast domains (for example HR and IT). A **trunk** carries several VLANs (802.1Q tags). **STP** (Spanning Tree Protocol) prevents loops. A **Layer 3 switch** can also route.

Wi-Fi standard	Name	Band	Max speed (theoretical)
802.11b	—	2.4 GHz	11 Mbps
802.11a / g	—	5 GHz / 2.4 GHz	54 Mbps

Wi-Fi standard	Name	Band	Max speed (theoretical)
802.11n	Wi-Fi 4	2.4 + 5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	~3.5 Gbps
802.11ax	Wi-Fi 6 / 6E	2.4 + 5 (+6) GHz	~9.6 Gbps

2.4 GHz = longer range, slower, more interference. **5 GHz** = faster, shorter range. **Wireless security**: WEP (broken) → WPA (TKIP) → **WPA2** (AES-CCMP) → **WPA3** (SAE, strongest). The **SSID** is the network name. Ethernet is IEEE 802.3, Wi-Fi is 802.11 and Bluetooth was 802.15.1.

🔐 PART 2 · CYBERSECURITY

Security

S

Protecting systems and data: goals, cryptography, identity, threats and defences. About 30% of Section C. Most questions ask you to name the right term for a description.

S1

CORE IDEAS

Security Fundamentals

C · Confidentiality
Only authorised people can read the data. *Tools: encryption, access control.*

I · Integrity
The data isn't altered without detection. *Tools: hashing, digital signatures, checksums.*

A · Availability
Systems are up when needed. *Tools: backups, redundancy, DDoS protection.*

Term	Meaning
AAA	**Authentication** (who you are) → **Authorization** (what you may do) → **Accounting** (logging what you did)
Non-repudiation	The sender can't deny sending something (digital signatures)
Threat / Vulnerability / Risk	A threat is a possible danger · a vulnerability is a weakness · risk = likelihood × impact of a threat exploiting a vulnerability
Exploit	Code or a technique that takes advantage of a vulnerability
Least privilege	Give only the minimum access needed
Defence in depth	Multiple layers of security controls
Zero Trust	"Never trust, always verify." Every request is authenticated, even inside the network

S2

SECRET WRITING

Cryptography & PKI

	Symmetric	Asymmetric (public-key)	Hashing
Keys	**One shared key**	**Public + private** key pair	No key
Reversible?	Yes (decrypt)	Yes (decrypt with the other key)	**No**: one-way

	Symmetric	Asymmetric (public-key)	Hashing
Speed	Fast (bulk data)	Slow (key exchange, signatures)	Very fast
Examples	**AES** (128/192/256-bit), DES (56-bit, weak), 3DES, Blowfish	**RSA**, ECC, Diffie–Hellman (key exchange), DSA	**SHA-256**, SHA-3 · MD5 (128-bit) and SHA-1 (160-bit) are broken
Use	Encrypting files, disks, sessions	Sharing keys, signing, certificates	Passwords (with a **salt**), integrity checks

Digital signature
Hash the message, then encrypt the hash with the sender's **private key**. The receiver decrypts it with the sender's **public key** and compares hashes. This proves **authenticity, integrity and non-repudiation**. To send a **secret** message instead, encrypt it with the **receiver's public key**.

PKI & certificates
A **Certificate Authority (CA)** issues **X.509 digital certificates** that bind a public key to an identity. Browsers trust root CAs. **TLS** (the successor to SSL) uses certificates plus asymmetric key exchange, then a **symmetric** session key for speed.

Encoding ≠ encryption ≠ hashing: Base64 and URL encoding only change the representation and anyone can reverse them. Encryption needs a key to reverse. Hashing can't be reversed at all.

S3

IDENTITY

Authentication & Access

Factor	Examples
Something you know	Password, PIN, security question
Something you have	Phone (OTP), smart card, hardware token
Something you are	Fingerprint, face, iris (biometrics)

MFA uses factors from **different** categories (a password plus an OTP). Two passwords are still single-factor. **2FA** = exactly two factors.

Concept	Meaning
SSO	Single Sign-On: log in once and access many apps
OAuth 2.0	**Authorization** framework ("allow this app to access my Google Drive") using access tokens
OpenID Connect	**Authentication** layer on top of OAuth ("Sign in with Google")
SAML	XML-based SSO between an identity provider and an enterprise app
Kerberos	Ticket-based authentication in Windows domains (port 88)
RBAC / ABAC	Access based on role (Admin, Viewer) / on attributes (department, time)
JWT	JSON Web Token: a signed token carrying user claims, common in APIs

S4

KNOW YOUR ENEMY

Malware & Attacks

Malware	Key trait	Malware	Key trait
Virus	Attaches to a host file. Spreads when the file runs	**Worm**	**Self-replicates** across networks, with **no host** and no user action
Trojan	Disguised as legitimate software	**Ransomware**	Encrypts files and demands payment (WannaCry)
Spyware	Secretly collects information	**Keylogger**	Records keystrokes (passwords)
Adware	Unwanted ads	**Rootkit**	Hides deep in the OS and keeps admin access
Botnet	A network of infected "zombie" devices (used for DDoS)	**Backdoor**	A secret way in that bypasses authentication
Logic bomb	Triggers on a condition or date	**Fileless**	Lives only in memory or scripts

Attack	How it works	Defence
Phishing	Fake email or site tricks the user into revealing data. **Spear phishing** = targeted · **Whaling** = executives · **Vishing** = voice · **Smishing** = SMS	Awareness training, email filtering, MFA
Social engineering	Manipulating people: **pretexting** (a made-up story), **baiting** (an infected USB), **tailgating/piggybacking** (following someone through a secure door), quid pro quo	Training, verification procedures
DoS / DDoS	Flooding a service (from many sources for DDoS, often a botnet): SYN flood, UDP flood	Rate limiting, CDN/scrubbing, redundancy
Man-in-the-Middle	Intercepting or altering traffic between two parties (fake Wi-Fi, **ARP spoofing**)	TLS/HTTPS, VPN, certificate checks
DNS spoofing / poisoning	Fake DNS answers redirect users to malicious sites	DNSSEC
SQL Injection	Malicious SQL in input: `' OR '1'='1`	**Parameterised queries**, input validation
XSS	Injected scripts run in other users' browsers	Output encoding, CSP
CSRF	Tricks a logged-in user's browser into sending an unwanted request	CSRF tokens, SameSite cookies
Brute force / dictionary	Trying many passwords. **Credential stuffing** reuses leaked passwords. **Password spraying** tries one common password on many accounts	Lockouts, MFA, strong unique passwords
Session hijacking	Stealing a session cookie or token	HTTPS, HttpOnly/Secure cookies, timeouts
Zero-day	Exploits a flaw before the vendor has a patch	Defence in depth, rapid patching

S5

BUILDING WALLS

Network Defences & Security Testing

Firewall type	How it filters
Packet-filtering (stateless)	Each packet by IP, port and protocol rules (L3/L4)
Stateful inspection	Tracks connection state and allows return traffic of established sessions
Proxy / application-level	Acts on the client's behalf and inspects application data (L7)
Next-Gen Firewall (NGFW)	Stateful filtering + app awareness + IPS + threat intelligence
WAF	Web Application Firewall: blocks SQLi, XSS and similar for websites

IDS vs IPS

IDS (Intrusion Detection) **detects and alerts** (passive). **IPS** (Intrusion Prevention) sits inline and **blocks**. Both can be signature-based or anomaly-based.

VPN

An encrypted tunnel over a public network. **Site-to-site** links offices, and **remote access** connects a user to the office. Protocols: **IPsec**, SSL/TLS VPN, WireGuard.

DMZ

A buffer zone between the internet and the internal LAN, for public servers (web, mail).

Honeypot

A decoy system that lures and studies attackers.

SIEM

Collects and correlates logs to detect incidents (Splunk, QRadar). A **SOC** is the team that monitors them.

Endpoint security

Antivirus/EDR, patching, encryption, least privilege, backups (the **3-2-1 rule**).

Vulnerability assessment vs penetration testing

A VA **finds and lists** weaknesses (scanners such as Nessus). A pen test **actively exploits** them to show the impact. **White box** = full knowledge · **black box** = none · **grey box** = partial. **Ethical hacking** is authorised testing.

OWASP Top 10

A list of the most critical web app risks, such as broken access control, cryptographic failures, injection, insecure design, security misconfiguration, vulnerable components, and authentication failures.

☁ PART 3 · CLOUD & DEVOPS

Cloud Computing & DevOps

K

How modern IT is delivered and built: cloud service and deployment models, virtualisation, scaling, SDLC and Agile, CI/CD, Git, Docker and Kubernetes. About 30% of Section C.

K1

THE CLOUD

Cloud Computing & Service Models

Cloud computing means on-demand delivery of computing resources (servers, storage, databases, software) over the internet with **pay-as-you-go** pricing. It shifts spending from **CapEx** (buying hardware) to **OpEx** (renting). NIST's **5 essential characteristics**: on-demand self-service · broad network access · resource pooling (multi-tenancy) · rapid elasticity · measured service.

Layer: who manages it?	On-premise	IaaS	PaaS	SaaS
Applications	You	You	You	Provider
Data	You	You	You	Provider*
Runtime & middleware	You	You	Provider	Provider
Operating system	You	You	Provider	Provider
Virtualisation, servers, storage, network	You	Provider	Provider	Provider

*In SaaS you still own your data and control who has access to it. This is the **shared responsibility model**.

Model	You get	Examples
IaaS	Virtual machines, storage, networks	AWS EC2, Azure Virtual Machines, Google Compute Engine
PaaS	A platform to deploy code. No OS management	Google App Engine, Heroku, Azure App Service, AWS Elastic Beanstalk
SaaS	Finished software in a browser	Gmail, Microsoft 365, Salesforce, Zoom, Dropbox
FaaS (serverless)	Run functions on events and pay per execution	AWS Lambda, Azure Functions, Google Cloud Functions

Deployment	Meaning	Example
Public	Shared provider infrastructure, open to anyone	AWS, Azure, GCP
Private	Dedicated to one organisation (on-premise or hosted)	A bank's internal cloud (OpenStack, VMware)
Hybrid	Private + public working together	Sensitive data on private, burst capacity on public
Community	Shared by organisations with common needs	Government agencies
Multi-cloud	Using several public providers	AWS + Azure, to avoid lock-in

K2

UNDER THE HOOD

Virtualisation, Scaling & Storage

Virtualisation

A **hypervisor** runs several virtual machines on one physical server.
Type 1 (bare-metal) runs directly on the hardware: VMware ESXi, Microsoft Hyper-V, KVM, Xen.
Type 2 (hosted) runs on top of an OS: VirtualBox, VMware Workstation.

	Virtual machine	Container
Contains	A full guest OS + app	App + dependencies only
Kernel	Its own	**Shares** the host kernel
Size / start	GBs / minutes	MBs / seconds
Isolation	Strong	Process-level
Tools	VMware, Hyper-V	Docker, Kubernetes

Scaling

Vertical (scale up) = a bigger machine (more CPU/RAM). **Horizontal (scale out)** = more machines. **Elasticity** = scaling automatically up *and down* with demand (auto-scaling).

High availability

Redundancy across **Availability Zones** (separate data centres within a **region**), load balancers and failover. SLAs are written as "nines" (99.99%).

CDN

Edge servers cache content near users for speed (CloudFront, Azure CDN, Cloudflare).

Storage type	How data is stored	Use	AWS / Azure / GCP
Object	Objects with metadata in flat buckets, accessed over HTTP	Backups, media, data lakes	S3 / Blob Storage / Cloud Storage
Block	Raw volumes, like a disk	VM disks, databases	EBS / Managed Disks / Persistent Disk
File	A shared hierarchical file system	Shared folders	EFS / Azure Files / Filestore

Service	AWS	Azure	Google Cloud
Virtual machines	EC2	Virtual Machines	Compute Engine
Serverless functions	Lambda	Functions	Cloud Functions
Managed SQL	RDS	SQL Database	Cloud SQL
NoSQL	DynamoDB	Cosmos DB	Firestore / Bigtable
Kubernetes	EKS	AKS	GKE
Private network	VPC	VNet	VPC
Identity & access	IAM	Entra ID (Azure AD)	Cloud IAM

K3

HOW SOFTWARE GETS BUILT

SDLC, Agile & DevOps

SDLC phases: Requirements → Design → Implementation (coding) → Testing → Deployment → Maintenance.

Model	Idea	Best for
Waterfall	Sequential phases with no going back	Fixed, clear requirements
V-Model	Waterfall with a matching test phase for each development phase	Safety-critical systems
Iterative / Incremental	Build in repeated cycles or pieces	Evolving products
Spiral	Iterations driven by **risk analysis**	Large, high-risk projects
Agile	Short iterations, customer feedback, working software over documentation	Changing requirements (most modern teams)

Scrum (an Agile framework)
Roles: Product Owner (owns the backlog and priorities) · Scrum Master (facilitator, removes blockers) · Developers.
Events: Sprint (1–4 weeks) · Sprint Planning · Daily Stand-up (15 minutes) · Sprint Review · Retrospective.
Artifacts: Product Backlog · Sprint Backlog · Increment. **Kanban** is a visual board with WIP limits and no fixed sprints.

DevOps
A culture plus practices that unite development and operations for fast, reliable delivery.
Lifecycle: **Plan → Code → Build → Test → Release → Deploy → Operate → Monitor** (an infinite loop).
CI: merge small changes often, with an automatic build and test on every commit. **Continuous Delivery**: always releasable, with a manual approval. **Continuous Deployment**: released automatically.

Stage	Popular tools	Stage	Popular tools
Version control	Git, GitHub, GitLab, Bitbucket	Containers	Docker
CI/CD	Jenkins, GitHub Actions, GitLab CI, Azure DevOps	Orchestration	Kubernetes
Build	Maven, Gradle, npm	Config management	Ansible, Chef, Puppet
Testing	Selenium, JUnit	Infrastructure as Code	Terraform, CloudFormation
Monitoring	Prometheus, Grafana, ELK stack	Artifact repository	Nexus, JFrog Artifactory

K4

HANDS-ON TOOLS

Git, Docker & Kubernetes

Git: the 4 areas

Working directory — **git add →** **Staging area (index)** — **git commit →** **Local repository** — **git push →** **Remote (GitHub)**

`git pull` = `git fetch` (download) + `git merge`. Git is a **distributed** VCS: every clone has the full history. SVN is centralised.

Command	Does	Command	Does
`git init`	Create a new repo	`git branch x`	Create a branch

Command	Does	Command	Does
`git clone <url>`	Copy a remote repo locally	`git checkout x` / `git switch x`	Switch branch
`git status`	Show changed and staged files	`git merge x`	Merge x into the current branch
`git add .`	Stage all changes	`git rebase`	Replay commits onto another base
`git commit -m "msg"`	Save a snapshot locally	`git stash`	Shelve uncommitted work
`git push`	Upload commits to the remote	`git log` / `git diff`	History / changes
`git fetch`	Download without merging	`git revert`	New commit that undoes an old one (safe)
`git pull`	Fetch + merge	`git reset`	Move HEAD back (can discard work)

Fork = your own server-side copy of someone else's repo. **Pull/merge request** = asking for your branch to be reviewed and merged. A **merge conflict** happens when two changes touch the same lines, so you fix them by hand. `.gitignore` lists files Git should ignore.

```
# Dockerfile
FROM python:3.12-slim     # base image
WORKDIR /app
COPY . .                  # copy source
RUN pip install -r req.txt # runs at BUILD
time
EXPOSE 8000               # documents the port
CMD ["python", "app.py"]  # default at RUN
time
```

Docker term	Meaning
Image	A read-only, layered template
Container	A **running instance** of an image
Dockerfile	The recipe used to build an image
Registry	Stores images (Docker Hub, ECR)
Volume	Persistent data outside the container
Compose	A YAML file that runs multi-container apps

Commands: `docker build -t app .` · `docker run -p 8000:8000 app` · `docker ps` (running containers) · `docker images` · `docker pull/push` · `docker stop` · `docker logs` · `docker exec -it id sh`.

Kubernetes object	Meaning
Cluster	A control plane plus **worker nodes** (machines)
Pod	The **smallest deployable unit**: one or more containers sharing a network and storage
Deployment	Declares the desired number of pod replicas, with rolling updates and self-healing
Service	A stable network endpoint / load balancer in front of pods
Namespace · **ConfigMap** · **Secret**	Logical isolation · configuration · sensitive values

NETWORKING

Practice Set 1 · Networking

S1–S8 are subnetting questions, **computed with Python's ipaddress library**. N1–N20 cover concepts, layers, devices, ports and protocols.

S1 Subnetting **How many usable host addresses are in a `/27` subnet?**

A 32
B 62
C 27
D 30

S2 Subnetting **What is the network address of host `192.168.10.77/26`?**

A 192.168.10.64
B 192.168.10.0
C 192.168.10.128
D 192.168.10.76

S3 Subnetting **What is the broadcast address of `172.16.5.130/25`?**

A 172.16.5.127
B 172.16.5.255
C 172.16.255.255
D 172.16.5.128

S4 Subnetting **What is the subnet mask for prefix `/28`?**

A 255.255.255.240
B 255.255.255.224
C 255.255.255.248
D 255.255.240.0

S5 Subnetting **What is the last usable host in `10.1.1.0/29`?**

A 10.1.1.8
B 10.1.1.6
C 10.1.1.7
D 10.1.1.5

S6 Subnetting **How many `/26` subnets fit in one `/24` network?**

A 6
B 2
C 4
D 64

S7 Subnetting **Which address is in the same subnet as `192.168.1.70/27`?**

A 192.168.1.90
B 192.168.1.30
C 192.168.1.60
D 192.168.1.100

S8 Subnetting **`172.20.14.9` is a:**

A Loopback address
B Multicast address
C Private IPv4 address
D Public IPv4 address

N1 Networking **In which topology is every device connected to every other device?**

A Ring
B Bus
C Mesh
D Star

N2 Networking **How many links does a full mesh of 6 devices need?**

A 6
B 30
C 12
D 15

N3 Networking **Communication that goes in both directions, but only one at a time (like a walkie-talkie), is:**

A Simplex
B Full duplex
C Multiplex
D Half duplex

N4 Networking **Which OSI layer does encryption, compression and data formatting?**

A Presentation (6)
B Application (7)
C Transport (4)
D Session (5)

N5 Networking **The data unit at the Data Link layer is called a:**

A Packet
B Frame
C Segment
D Bit

N6 Networking **Which device separates BROADCAST domains?**

A Repeater
B Switch
C Hub
D Router

N7 Networking **Which protocol finds the MAC address for a known IP address?**

A DHCP
B ICMP
C ARP
D DNS

N8 Networking **A MAC address is how long?**

A 48 bits
B 64 bits
C 128 bits
D 32 bits

N9 Networking **IPv6 addresses are how long?**

A 128 bits
B 32 bits
C 256 bits
D 64 bits

N10 Networking **Which of these does IPv6 NOT have?**

A Multicast addresses
B Unicast addresses
C Broadcast addresses
D Anycast addresses

N11 Networking **What is the correct order of the TCP 3-way handshake?**

A FIN → ACK → SYN
B SYN → ACK → SYN-ACK
C ACK → SYN → FIN
D SYN → SYN-ACK → ACK

N12 Networking **Which transport protocol do DNS queries normally use?**

A TCP (port 80)
B UDP (port 67)
C TCP (port 25)
D UDP (port 53)

N13 Networking **SSH runs on which port?**

A 23
B 21
C 22
D 25

N14 Networking **Which protocol is used to RETRIEVE email while keeping it synced on the server?**

A POP3
B IMAP
C SMTP
D SNMP

N15 Networking **HTTP status 301 means:**

A Moved permanently (redirect)
B Server error
C Unauthorized
D Created

N16 Networking **Which HTTP method is meant for creating a new resource?**

A POST
B DELETE
C GET
D HEAD

N17 Networking **Which DNS record maps a domain name to an IPv4 address?**

A AAAA
B A
C MX
D CNAME

N18 Networking **The DHCP process in order is:**

A Discover, Offer, Request, Acknowledge
B Discover, Request, Offer, Acknowledge
C Request, Offer, Discover, Acknowledge
D Offer, Discover, Acknowledge, Request

N19 Networking **RIP considers a route unreachable at a hop count of:**

A 16
B 255
C 8
D 15

N20 Networking **Which Wi-Fi security protocol is the newest and strongest?**

A WPA2
B WPA
C WEP
D WPA3

Set 1 · Answer key

Q	Ans	Correct	Why
S1	D	30	Host bits = 32 − 27 = 5, and $2^5 - 2 = 30$. The network and broadcast addresses aren't usable.
S2	A	192.168.10.64	A /26 makes blocks of 64: .0, .64, .128 and .192. 77 falls in the .64 block.

Q	Ans	Correct	Why
S3	**B**	172.16.5.255	A /25 makes blocks of 128. 130 is in the .128 block, so its broadcast is the last address, .255.
S4	**A**	255.255.255.240	28 ones: 255.255.255 + 11110000_2 (240).
S5	**B**	10.1.1.6	A /29 has 8 addresses: .0 (network), .1–.6 (hosts) and .7 (broadcast).
S6	**C**	4	You borrow 2 bits (26 − 24), and $2^2 = 4$ subnets of 64 addresses each.
S7	**A**	192.168.1.90	A /27 makes blocks of 32. .70 is in .64–.95, and only .90 falls in that range.
S8	**C**	Private IPv4 address	172.16.0.0 to 172.31.255.255 is a private range (RFC 1918).
N1	**C**	Mesh	A full mesh of n devices needs n(n−1)/2 links. It's very reliable but expensive.
N2	**D**	15	n(n−1)/2 = 6 × 5 / 2 = 15.
N3	**D**	Half duplex	Simplex is one direction only (a keyboard or TV broadcast). Full duplex is both directions at once (a phone call).
N4	**A**	Presentation (6)	Layer 6 translates the data format.
N5	**B**	Frame	L4 = segment, L3 = packet, L2 = frame, L1 = bits.
N6	**D**	Router	Switches separate collision domains, routers separate broadcast domains, and hubs separate neither.
N7	**C**	ARP	ARP (Address Resolution Protocol). DNS resolves names to IPs.
N8	**A**	48 bits	48 bits, written as 6 hex pairs. The first 24 bits (the OUI) identify the manufacturer.
N9	**A**	128 bits	Written as 8 groups of 4 hex digits. :: compresses consecutive zero groups, and can be used only once.
N10	**C**	Broadcast addresses	IPv6 replaces broadcast with multicast.
N11	**D**	SYN → SYN-ACK → ACK	Connection teardown uses FIN and ACK (4 steps).
N12	**D**	UDP (port 53)	DNS uses UDP 53 for normal queries and TCP 53 for zone transfers and large responses.
N13	**C**	22	21 = FTP, 22 = SSH, 23 = Telnet (unencrypted), 25 = SMTP.
N14	**B**	IMAP	SMTP sends mail. POP3 downloads it (and usually deletes it from the server). IMAP keeps it synced on the server.
N15	**A**	Moved permanently (redirect)	401 = unauthorized, 5xx = server errors, 201 = created.
N16	**A**	POST	GET reads, POST creates, PUT replaces, PATCH partially updates, DELETE removes.
N17	**B**	A	AAAA is for IPv6, MX for mail servers, and CNAME for aliases.
N18	**A**	Discover, Offer, Request, Acknowledge	Remember DORA.
N19	**A**	16	RIP's maximum usable hop count is 15. OSPF is link-state, and BGP routes between autonomous systems (the internet).
N20	**D**	WPA3	WEP is broken. WPA2 uses AES, and WPA3 adds SAE.

SECURITY

Practice Set 2 · Security

X1 Security **Ensuring data is not altered in transit is which part of the CIA triad?**

A Integrity
B Confidentiality
C Availability
D Authentication

X2 Security **Verifying WHO a user is, is called:**

A Accounting
B Authorization
C Authentication
D Auditing

X3 Security **A password plus an OTP sent to your phone is an example of:**

A Biometrics
B Single sign-on
C Multi-factor authentication
D Role-based access

X4 Security **Which algorithm is asymmetric?**

A RSA
B 3DES
C AES
D DES

X5 Security **Hashing is used to store passwords because it is:**

A Compressed
B One-way (it can't be reversed)
C Fast to decrypt
D Symmetric

X6 Security **Base64 is:**

A A digital signature
B Strong encryption
C A hash function
D Encoding, not encryption

X7 Security **A digital signature is created with the sender's:**

A Password
B Public key
C Private key
D Session key

X8 Security **Malware that spreads across networks BY ITSELF, without a host file, is a:**

A Virus
B Trojan
C Rootkit
D Worm

X9 Security **Malware disguised as useful software is a:**

A Worm
B Botnet
C Trojan
D Adware

X10 Security **A targeted phishing attack on a company's CEO or senior executives is called:**

A Whaling
B Vishing
C Baiting
D Smishing

X11 Security **An attacker follows an employee through a secure door without a badge. This is:**

A Pretexting
B Sniffing
C Tailgating
D Spoofing

X12 Security **Thousands of infected devices flood a website so real users can't reach it. This is:**

A Brute force
B SQL injection
C DDoS
D MITM

X13 Security **Injecting malicious JavaScript into a web page that other users view is:**

A Cross-Site Scripting (XSS)
B SQL injection
C Session fixation
D CSRF

X14 Security **The best defence against SQL injection is:**

A A bigger database
B Strong passwords
C Parameterised queries / prepared statements
D Using HTTPS

X15 Security **Which device DETECTS and also BLOCKS malicious traffic inline?**

A IPS
B Proxy cache
C IDS
D Hub

X16 Security **A network zone holding public-facing servers, isolated from the internal LAN, is a:**

A VLAN trunk
B Honeypot
C VPN
D DMZ

X17 Security **A firewall that tracks the state of connections is:**

A Stateful
B Circuit-free
C Static NAT
D Packet-filtering (stateless)

X18 Security **Testing a system with FULL knowledge of its internals is:**

A Grey-box testing
B Black-box testing
C White-box testing
D Fuzzing only

Set 2 · Answer key

Q	Ans	Correct	Why
X1	A	Integrity	Hashes and digital signatures protect integrity.
X2	C	Authentication	Authorization decides what they can do. Accounting logs what they did (together, AAA).
X3	C	Multi-factor authentication	Something you KNOW plus something you HAVE.
X4	A	RSA	RSA uses a public/private key pair. The others use one shared key.
X5	B	One-way (it can't be reversed)	Add a unique salt to each password to defeat rainbow tables.

Q	Ans	Correct	Why
X6	**D**	Encoding, not encryption	Anyone can decode Base64. It only changes the representation.
X7	**C**	Private key	Anyone verifies it with the sender's public key. That gives authenticity, integrity and non-repudiation.
X8	**D**	Worm	A virus needs a host file and a user action.
X9	**C**	Trojan	Named after the Trojan horse.
X10	**A**	Whaling	Vishing uses voice calls and smishing uses SMS.
X11	**C**	Tailgating	A physical social-engineering attack.
X12	**C**	DDoS	Distributed Denial of Service, often launched from a botnet.
X13	**A**	Cross-Site Scripting (XSS)	Prevent it with output encoding and input validation.
X14	**C**	Parameterised queries / prepared statements	They keep data separate from the SQL code.
X15	**A**	IPS	An IDS only detects and alerts. An IPS prevents.
X16	**D**	DMZ	Demilitarised zone. Web and mail servers often sit there.
X17	**A**	Stateful	A WAF protects web apps (Layer 7). A next-generation firewall (NGFW) adds application awareness and IPS.
X18	**C**	White-box testing	Black box means no knowledge. Grey box means partial knowledge.

CLOUD & DEVOPS

Practice Set 3 · Cloud & DevOps

C1 Cloud & DevOps **Which is NOT one of NIST's 5 essential cloud characteristics?**

A Measured service
B Fixed long-term hardware contracts
C On-demand self-service
D Rapid elasticity

C2 Cloud & DevOps **In PaaS, the CUSTOMER manages:**

A Applications and data
B Networking hardware
C The hypervisor
D Physical servers

C3 Cloud & DevOps **AWS Lambda / Azure Functions are examples of:**

A SaaS
B FaaS (serverless)
C IaaS
D On-premise

C4 Cloud & DevOps **Google Workspace, Salesforce and Zoom are examples of:**

A PaaS
B FaaS
C SaaS
D IaaS

C5 Cloud & DevOps **Moving from buying servers to paying monthly for cloud usage shifts spending from:**

A CapEx to CapEx
B OpEx to OpEx
C CapEx to OpEx
D OpEx to CapEx

C6 Cloud & DevOps **Adding MORE servers to handle load is:**

A Horizontal scaling (scale out)
B Virtualisation
C Caching
D Vertical scaling (scale up)

C7 Cloud & DevOps **VMware ESXi and Microsoft Hyper-V are:**

A Container engines
B Type 2 (hosted) hypervisors
C Type 1 (bare-metal) hypervisors
D Load balancers

C8 Cloud & DevOps **Compared with VMs, containers:**

A Need a type 1 hypervisor
B Each include a full guest OS
C Can't run on Linux
D Share the host OS kernel and start faster

C9 Cloud & DevOps **Amazon S3 / Azure Blob Storage are examples of:**

A Object storage
B Block storage
C Relational databases
D File storage

C10 Cloud & DevOps **A network of edge servers that caches content close to users is a:**

A VPC
B VPN
C CDN
D NAS

C11 Cloud & DevOps **Which SDLC model completes each phase fully before the next, with no going back?**

A Iterative
B Spiral
C Waterfall
D Agile

C12 Cloud & DevOps **In Scrum, a fixed time-box (usually 1–4 weeks) to deliver an increment is a:**

A Stand-up
B Backlog
C Sprint
D Epic

C13 Cloud & DevOps **In Continuous Integration, every code change:**

A Is merged often and automatically built and tested
B Must be emailed to the tester
C Is reviewed only at the end of the project
D Goes straight to production without tests

C14 Cloud & DevOps **Which Git command moves changes into the staging area?**

A git stash
B git commit
C git push
D git add

C15 Cloud & DevOps `git pull` **is equivalent to:**

A git fetch + git merge
B git add + git commit
C git clone + git push
D git branch + git checkout

C16 Cloud & DevOps **In a Dockerfile, which instruction sets the BASE image?**

A EXPOSE
B RUN
C FROM
D CMD

C17 Cloud & DevOps **The smallest deployable unit in Kubernetes is a:**

A Node
B Namespace
C Cluster
D Pod

C18 Cloud & DevOps **Writing infrastructure as version-controlled code (for example with Terraform) is called:**

A Containerisation
B Virtualisation
C Serverless
D Infrastructure as Code (IaC)

Set 3 · Answer key

Q	Ans	Correct	Why
C1	B	Fixed long-term hardware contracts	The 5 are: on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service.
C2	A	Applications and data	The provider manages everything up to the runtime and OS.
C3	B	FaaS (serverless)	You upload functions, they run on events, and you pay per execution.
C4	C	SaaS	Ready-to-use software delivered over the internet.
C5	C	CapEx to OpEx	CapEx is capital expenditure (upfront). OpEx is operational expenditure (pay as you go).

Q	Ans	Correct	Why
C6	A	Horizontal scaling (scale out)	Vertical scaling means a bigger machine.
C7	C	Type 1 (bare-metal) hypervisors	Type 2 hypervisors run on a host OS (VirtualBox, VMware Workstation).
C8	D	Share the host OS kernel and start faster	This makes them lightweight and portable.
C9	A	Object storage	Block storage is like a disk (EBS). File storage is a shared file system (EFS).
C10	C	CDN	Examples: CloudFront, Azure CDN, Cloudflare.
C11	C	Waterfall	Spiral is risk-driven. Agile works in short iterations.
C12	C	Sprint	Roles: Product Owner, Scrum Master, Developers.
C13	A	Is merged often and automatically built and tested	CD (Continuous Delivery/Deployment) automates the release.
C14	D	git add	Working directory → (add) → staging → (commit) → local repo → (push) → remote.
C15	A	git fetch + git merge	git fetch only downloads and doesn't merge.
C16	C	FROM	RUN runs at build time, CMD is the default command at run time, and EXPOSE documents a port.
C17	D	Pod	A pod holds one or more containers. A node is a machine. A cluster is a set of nodes.
C18	D	Infrastructure as Code (IaC)	Ansible, Chef and Puppet handle configuration management.

GET EXAM-READY

5-Day Plan & Cheat Sheet

Day	Plan (about 1.5 hours)
Day 1	N1–N3: topologies, the OSI table (layer, device, PDU, protocols), devices. Make flashcards.
Day 2	N4–N7: IP classes and private ranges, **subnetting drills** (20 problems), TCP vs UDP, the ports table, HTTP codes, DNS and DHCP.
Day 3	S1–S5: CIA and AAA, the crypto comparison, MFA factors, the malware and attack tables, firewalls, IDS/IPS, VPN, DMZ.
Day 4	K1–K4: the responsibility matrix, examples of each model, virtualisation, scaling, storage types, Scrum, CI/CD, Git, Docker and Kubernetes.
Day 5	All three practice sets, timed at about 40 seconds per question. Re-read the tables for every question you get wrong.

Networking

- OSI 7→1: App, Pres, Sess, Trans, Net, DL, Phys
- PDU: data → segment → packet → frame → bits
- Hub L1 · switch L2 (MAC) · router L3 (IP)
- Router separates broadcast domains
- Private: 10/8 · 172.16–31 · 192.168/16 · loopback 127
- Hosts = $2^h - 2$ · /26 = 62 · /27 = 30 · /28 = 14
- TCP: SYN, SYN-ACK, ACK · UDP: fast, no guarantee
- 21 FTP · 22 SSH · 23 Telnet · 25 SMTP · 53 DNS
- 67/68 DHCP · 80 HTTP · 443 HTTPS · 3389 RDP
- DHCP = DORA · DNS: A, AAAA, CNAME, MX
- 4xx client error · 5xx server error · 404 / 403 / 500
- IPv6 128-bit, no broadcast · MAC 48-bit · ARP: IP → MAC

Security

- CIA: Confidentiality, Integrity, Availability
- AuthN = who · AuthZ = what · Accounting = log
- Symmetric AES · asymmetric RSA · hash SHA-256
- MD5 and SHA-1 are broken · salt your hashes
- Sign with your private key · encrypt with the receiver's public key
- MFA = know + have + are (different factors)
- Worm self-spreads · virus needs a host · trojan disguised
- Whaling = executives · vishing = voice · smishing = SMS
- SQLi → parameterised queries · XSS → encode output
- IDS detects · IPS blocks · WAF protects web apps
- DMZ for public servers · VPN = encrypted tunnel
- White / black / grey box = full / no / partial knowledge

Cloud & DevOps

- IaaS EC2 · PaaS App Engine · SaaS Gmail · FaaS Lambda
- PaaS: you manage the app and data only
- Public · Private · Hybrid · Community · Multi-cloud
- CapEx → OpEx · NIST's 5 characteristics
- Type 1 hypervisor ESXi/Hyper-V · Type 2 VirtualBox
- Containers share the kernel · VMs have a full OS
- Scale up = bigger · scale out = more machines
- Object S3 · block EBS · file EFS
- Waterfall sequential · Spiral risk · Agile iterations
- Scrum: PO, SM, Devs · sprint 1–4 weeks
- add → commit → push · pull = fetch + merge
- FROM base · container = running image · pod is the smallest

Disclaimer: This is an independent preparation guide for informational purposes only, not affiliated with or endorsed by Accenture or any vendor named. Product names belong to their owners. Wi-Fi speeds are theoretical maximums. All questions are original practice questions.